

Agent42

Hyper-Adaptive Smart Service Agent for the Nation

MOEI x 42 Abu Dhabi | AgentEra Hackathon | Challenge 3

Team Agent42

خدمة حكومة الإمارات بذكاء يستحقه المواطن

Meet Fatima Al Mansouri — Ajman — SZHP Housing Case

TODAY

4 months behind on SZHP payments after a medical emergency

Calls MOEI hotline — 40-minute wait, transferred twice

Re-explains herself from scratch on every channel

Tries WhatsApp — no memory of the call

5 DAYS | 4 CHANNELS | 1 HOUR WAIT | No answer



WITH Agent42

Sends one WhatsApp — Hassan already knows her

SZHP rules engine: rescheduling plan in 8 seconds

Case created automatically. SLA clock starts.

Proactive status update sent before she has to ask

90 SECONDS | 1 CONVERSATION | Never repeats

The Problem Is Bigger Than Fatima

174

federal services
at MOEI

4

channels per service
web · voice · WhatsApp · mobile

0

shared citizen context
between any of them — today

Siloed CRMs · Manual handoffs · No cross-channel memory · Reactive, not proactive

UAE National AI Strategy 2031 · AED 335 billion digital economy target

No team here is solving the cross-channel handoff. We are.

One Brain. Every Channel.

WhatsApp

Voice

Web / Chat

Mobile

V V V V

FastAPI Channel Gateway | correlation-ID | language detection | PII redaction at entry

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LangGraph Supervisor | Router > Sentiment > Memory > Guardrails > Dispatcher > Escalation > Composer

V

Housing
(SZHP)

Energy
(Tariffs)

Transport &
Maritime

Knowledge
(FTS / RAG)

CRM
(Profile)

PostgreSQL (system of record) | Redis (cross-channel session) | Langfuse (tracing + audit) | Postgres FTS (bilingual RAG)

LIVE DEMO

Fatima's journey — four channels — one memory — in real time.

Web Chat > WhatsApp > Proactive Update > Human Escalation

Router

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Sentiment

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Memory

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Guardrails

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Dispatcher

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Escalation

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Composer

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Persist

Multi-agent specialist dispatch

Housing · Energy · Transport · Maritime · Infrastructure — each with dedicated tools and deterministic sub-logic.

Predictive escalation engine

6 CRM signals fused (anger · SLA breach · reopened · repeat escalator · critical · VIP). Escalates on ≥ 2 . Predicts before citizen complains.

Deterministic policy engine

SZHP rules in Python, not LLM. Every decision is citable: 'SZHP-R3.1: arrears ≥ 3 months, eligible for hardship rescheduling.' Right-to-explanation built in.

Full PDPL audit trail

Every node input/output captured to Langfuse. Every decision persisted to Postgres. Citizen-accessible. Replayable for any auditor.

Federal-Grade. Built for Production.

25 pts

PDPL Compliant

Art. 4 Consent | Art. 5 Purpose limitation | Art. 6 UAE residency | Art. 7 Access
| Art. 8 Security | Art. 9 Breach SOP

PII + Bias Guards

Emirates ID / IBAN / mobile / email redacted at gateway AND composer.
Nationality / gender / religion bias detector built in.

UAE PASS Identity

Synthetic identities in demo; real UAE PASS sandbox in Phase 1 pilot. Verified citizen flow — no guessing who Fatima is.

Audit Trail UI

Every decision clickable by the citizen or any auditor. PDPL Art. 7 right-to-explanation: exact rule citations, not LLM post-hoc rationalisation.

Data Residency

Azure UAE North / Core42 (G42 Cloud). Fully containerised — on-prem deployable if TDRA requires. Data never leaves UAE.

Provider Failover

Groq > Claude/OpenAI > Gemini. Automatic cascade. No single-vendor lock-in. Zero hard-fails for the citizen.

Arabic-First. Not an Afterthought.

Jais-Family-30B

Khaliji Arabic dialect — purpose-built for Gulf Arabic, not translated from English.

Native RTL UI

Right-to-left is first-class: web, chat, and mobile all render RTL natively. Not a CSS patch.

Code-switching

Arabic + English in the same turn handled cleanly — because citizens actually talk that way.

Bilingual RAG

Postgres FTS knowledge base indexed in both languages. Ask in Arabic, get Arabic-cited results.

"خدمة حكومة الإمارات بذكاء يستحقه المواطن. وكيل42"

No Lock-In. No Single Point of Failure.

LangGraph + Langfuse

Apache-2.0 / MIT. No proprietary orchestration. No vendor owns the graph.

Provider-agnostic LLM cascade

Groq | Cerebras | Claude (Anthropic) | OpenAI | Jais | Gemini — swap any provider in one config line. Graph logic unchanged.

\$0 RAG on Postgres FTS

Bilingual full-text search today. pgvector when scale demands. No vector-DB vendor required at pilot volume.

Graceful degradation everywhere

LLM down -> fallback model. Model missing -> heuristic. Guardrail error -> safe default. The citizen never sees a hard fail.

Containerised for UAE sovereignty

Deploy on Azure UAE North, Core42, or any on-prem cluster. Data residency is a deployment decision, not a rewrite.

What This Means for Citizens — and for MOEI

60%

self-service
deflection rate

< 90s

housing triage
(was 5+ minutes)

< 5s

first meaningful
reply

4.5/5

target CSAT
by Phase 3

174 services x 60% deflection x 5-min baseline AHT = millions of citizen-hours returned annually

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Cost to run: < AED 1.00 per conversation | AED 25K/month infrastructure at pilot volume

We Don't Need 3 Years. We Need 90 Days.

Days 1–30

Production Integration

UAE PASS sandbox — real verified identity

Azure UAE North / Core42 deployment

Meta WhatsApp Cloud API under MOEI account

PDPL DPIA signed by MOEI Data Protection Officer

Days 31–60

Shadow Mode Evaluation

Hassan reads every real citizen turn — never replies

Daily: Hassan recommendation vs. human decision

Target: $\geq 80\%$ agreement on 5,000-turn evaluation set

Cost target: $< \text{AED } 1.00$ per conversation confirmed

Days 61–90

Limited GA Launch

Live: housing arrears triage as default first-touch

EnergyAgent + TransportAgent workers go live

Citizen audit-trail UI in production

60% deflection rate measured against baseline

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UAE government service with the intelligence its citizens deserve.